

Diagnostic Evaluation Report

1. Framework Information

- **Framework:** X-RAG Diagnostic Framework (v1.0)
- **Trace ID:** `eede0799-8f8f-4d3a-b2a4-42fa2eb5a985`
- **Timestamp:** 2026-07-18T04:49:39.067444Z
- **Status:** COMPLETED

2. Executive Summary

- **Question:** What is the punishment for murder?
- **Answer:** According to the context provided, the punishment for murder is death or imprisonment for life, which means t
- **Health Score:** 0.67
- **Primary Issue:** UNKNOWN
- **Claims Supported:** 2 / 3
- **Summary:** Analysis completed.

3. Pipeline Overview

Stage	Status	Confidence	Observation
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CORPUS	PASS	0.9	Corpus contains sufficient relevant content, evidenced by at least one supported claim.
RETRIEVER	PASS	0.95	At least one retrieved chunk provides supporting evidence for a verified claim.
CHUNKING	UNKNOWN	1.0	Deterministic chunk-boundary evidence is currently unobservable.
GENERATOR	PASS	0.95	All generated claims are fully supported by evidence.
GROUNDING	PASS	0.9	No claims are contradicted by retrieved evidence.

4. Evaluation Metrics

- **Retrieved Chunks:** 5
- **Verified Claims:** 3 (Supported: 2, Partial: 0, Unsupported: 0, Contradicted: 0)
- **Grounding Score:** 0.67
- **Evidence Coverage:** 0.60
- **Average Entailment:** 0.67
- **Latencies:** Retrieval (1040.4ms) | Generation (1329.9ms) | Verification (286696.4ms)

5. Evidence Analysis

- **[a67e485d-e7a3-48ca-ad66-b04718d25243]** (SUPPORTED): The punishment for murder is death or imprisonment for life.
 - *Evidence (Chunk f69e6aef-3594-43c8-86e9-415389ec638f):* If B had murdered Z, he would have been subject to the punishment of death or imprisonment for life. Therefore, A is liable to imprisonment for a term which may extend to seven years and also to a fine; and if any hurt be done to Z in consequence of the abetment, he will be liable to imprisonment for a term which may extend to fourteen years, and to fine.
- **[79de8640-7df5-49c3-bbb7-c1e8fe7e5570]** (SUPPORTED): Imprisonment for life means the remainder of the person's natural life.
 - *Evidence (Chunk 63b53cdc-7519-44a9-b89f-c58baebf570b):* (2) When any person offending under sub-section (1) is under sentence of imprisonment for life, he may, if hurt is caused, be punished with death or with imprisonment for life, which shall mean the remainder of that person's natural life.
- **[36086a8b-3e8a-46d7-ac27-8143052de195]** (NOT_VERIFYABLE): This information is detailed in Chunk-ID [63b53cdc-7519-44a9-b89f-c58baebf570b].
 - *Evidence (Chunk 5be11998-52fb-42a5-bfea-bf5a90a58ef8):* Explanation.--For the purposes of this section the expression "Government" means,--
 - (a) in cases where the sentence is a sentence of death or is for an offence against any law relating to a matter to which the executive power of the Union extends, the Central Government; and
 - (b) in cases where the sentence (whether of death or not) is for an offence against any law relating to a matter to which the executive power of the State extends, the Government of the State within which the offender is sentenced.

6. Root Cause Analysis

- **Primary Cause:** UNKNOWN
- **Confidence:** 0.93
- **Reasoning:**
 - Starting root cause analysis for trace eede0799-8f8f-4d3a-b2a4-42fa2eb5a985.
 - Assumption Passed at CORPUS. Moving to next stage.

- Assumption Passed at RETRIEVER. Moving to next stage.
- Skipping CHUNKING: State is UNKNOWN. Insufficient evidence.
- Assumption Passed at GENERATOR. Moving to next stage.
- Assumption Passed at GROUNDING. Moving to next stage.
- No failures detected. Pipeline is healthy or evidence is purely UNKNOWN.

7. Corrective Action Plan

8. Evidence Traceability

- **Primary Cause:** UNKNOWN
- **Derived From:** RootCauseAnalysis
- **Pipeline Stage:** UNKNOWN
- **Supporting Claims:** a67e485d-e7a3-48ca-ad66-b04718d25243, 79de8640-7df5-49c3-bbb7-c1e8fe7e5570, 36086a8b-3e8a-46d7-ac2
- **Supporting Chunks:** f69e6aef-3594-43c8-86e9-415389ec638f, 63b53cdc-7519-44a9-b89f-c58baebf570b, 5be11998-52fb-42a5-bf6

9. Overall Assessment

- **Major Strength:** N/A
- **Major Weakness:** UNKNOWN
- **Next Priority:** Monitor pipeline
- **Recommendation:** Requires architectural adjustments.

10. Framework Footer

- **Generated By:** X-RAG Diagnostic Framework
- **Artifact Version:** 1.0

APPENDIX

Artifacts Generated

✓ RAGTrace

artifacts/rag_traces/eede0799-8f8f-4d3a-b2a4-42fa2eb5a985.json

✓ ClaimSet

artifacts/claim_sets/eede0799-8f8f-4d3a-b2a4-42fa2eb5a985.json

✓ VerificationResults

artifacts/verification/eede0799-8f8f-4d3a-b2a4-42fa2eb5a985.json

✓ PipelineStateMatrix

artifacts/pipeline_state_matrix/eede0799-8f8f-4d3a-b2a4-42fa2eb5a985.json

✓ RootCauseAnalysis

artifacts/root_cause_analysis/eede0799-8f8f-4d3a-b2a4-42fa2eb5a985.json

✓ CorrectiveActionPlan

artifacts/corrective_action_plan/eede0799-8f8f-4d3a-b2a4-42fa2eb5a985.json

✓ DiagnosticEvaluationReport

artifacts/reports/eede0799-8f8f-4d3a-b2a4-42fa2eb5a985.json